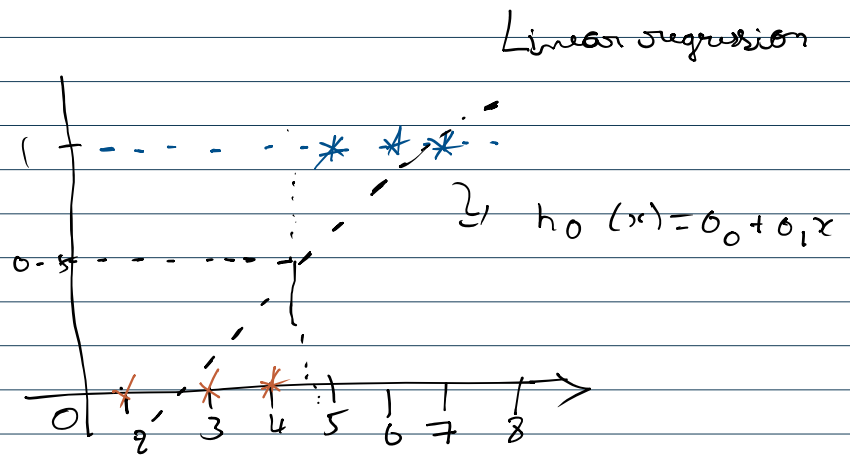
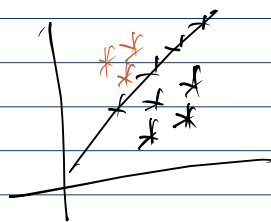


In **statistics**, a **logistic model** (or **logit model**) is a **statistical model** that models the **log-odds** of an event as a **linear combination** of one or more **independent variables**.

Dataset	I/p	O/p	
	Study hours	Result	study how → Pass/Fail
175	2	F	
20	3	F	
90	4	F	
	5	P	
	6	P	
	7	P	
	12	P	



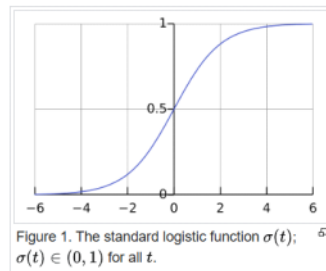
Background [\[edit\]](#)

Definition of the logistic function [\[edit\]](#)

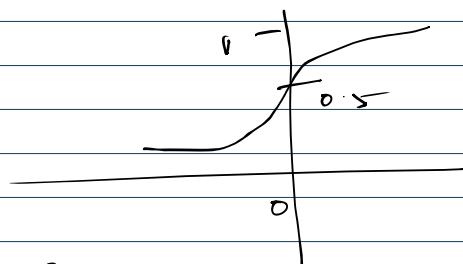
An explanation of logistic regression can begin with an explanation of the standard **logistic function**. The logistic function is a **sigmoid function**, which takes any **real** input t , and outputs a value between zero and one.^[2] For the logit, this is interpreted as taking input **log-odds** and having output **probability**. The **standard** logistic function $\sigma: \mathbb{R} \rightarrow (0, 1)$ is defined as follows:

$$\sigma(t) = \frac{e^t}{e^t + 1} = \frac{1}{1 + e^{-t}}$$

A graph of the logistic function on the t -interval $(-6, 6)$ is shown in Figure 1.



Sigmoid Activation



$$h_0(x) = \theta_0 + \theta_1 x$$

$$\frac{1}{1 + e^{-x}} \quad x=2 \quad \sigma(2) \approx 0.88$$

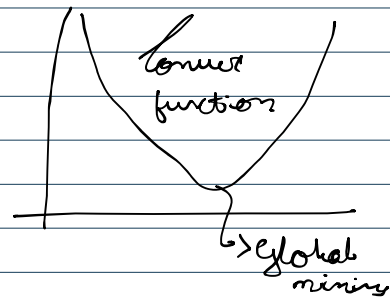
0-1

$$= \sigma(\theta_0 + \theta_1 x) \quad \text{where } \sigma = \frac{1}{1+e^{-z}}$$

$$h_0(x) = \frac{1}{1+e^{-z}} \quad (z = \theta_0 + \theta_1 x)$$

Linear Regression

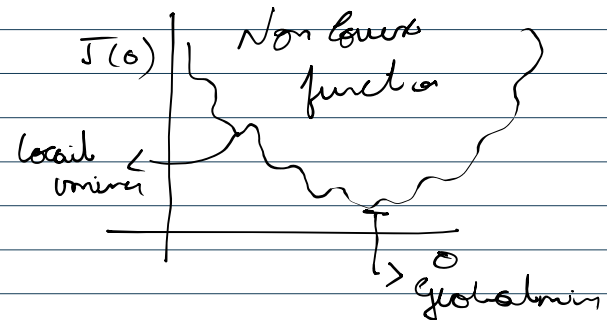
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$



Logistic Regression cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 \quad \frac{1}{1+e^{-z}}$$

Sigmoid



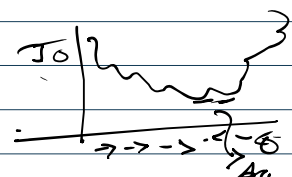
Log loss

$$\text{cost}(h_0(x^{(i)}), y^{(i)}) = \begin{cases} -\log(h_0(x)) & y=1 \\ -\log(1-h_0(x)) & y=0 \end{cases}$$

$$J(\theta_0, \theta_1) = -\frac{1}{2m} \sum_{i=1}^m (y^{(i)} \log(h_0(x^{(i)})) + (1-y^{(i)}) \log(1-h_0(x^{(i)})))$$

$\Rightarrow$
gradient descent

Convergen Alg



Repeat

{

$$\theta_j \approx \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}

Performance metrics

- i) Confusion matrix
- ii) Accuracy
- iii) Precision
- iv) Recall
- v) F beta score
- vi) ROC-AUC Curve

① Confusion matrix

	1	0
1	1+1 =3	1+1 =2
0	1+1 =2	1

3	2
2	1

Rough Dataset

Input x_1	Input x_2	True output y	Predicted output $\hat{y}$
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	0
-	-	0	1
-	-	1	0
-	-	0	0
-	-	1	1

	1	0
1	TP	FP
0	FN	TN

True Positive
False Positive
False Negative
True Negative

$$\text{② Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

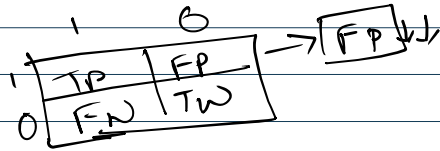
$$= \frac{3+1}{3+2+2+1} = \frac{4}{8} = \frac{1}{2} = 0.5 = 50\%$$

$$\text{③ Precision} = \frac{TP}{TP + FP} \quad \left\{ \begin{array}{l} \text{out of all actual values how many} \\ \text{are correctly predicted} \end{array} \right.$$

$$\text{④ Recall} = \frac{TP}{TP + FN} \quad \left\{ \begin{array}{l} \text{out of all the predicted value how} \\ \text{many are correctly predicted} \end{array} \right.$$

Use Case ①

Spam classification



Mail Spam ☐ → spam } ok
True Spam

Mail → ☐ → not spam ①
spam ②

Use Case ②

stud Pass / Fail

stud med → Pass } 10K
True → pass

Pass 0
True Fail 1 → FN ↓↓

F-beta score : $(1+\beta^2) \frac{\text{Precision} * \text{Recall}}{\text{precision} + \text{Recall}}$

① If FP & FN are both 0
 $\beta = 1$

$$F\text{-beta score} = (1+1)^2 * \frac{P * R}{P + R}$$

② If FP is more imp
 $\beta = 0.5$

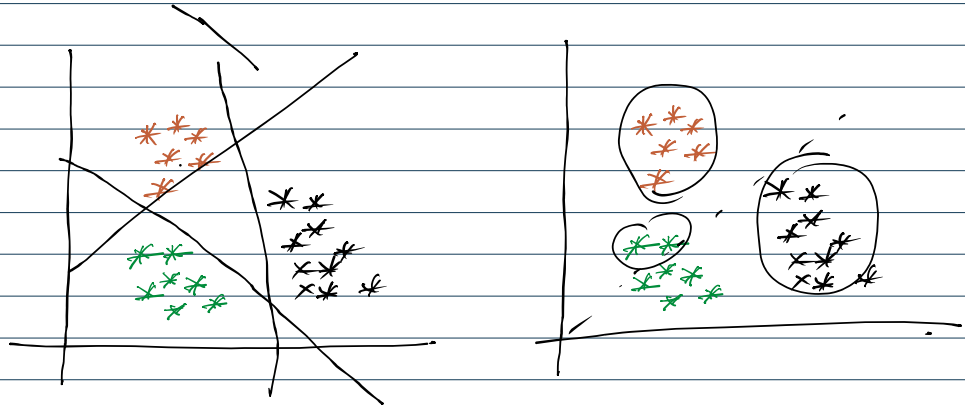
$$F_1 = (1+0.25) * \left[\frac{P * R}{P + R} \right]$$

③ FN ↑↑

$$\beta = 2$$

$$F_1 = (1+4) * \left[\frac{P * R}{P + R} \right]$$

Logistic Regression



x_1	x_2	x_3	O/P	O_1	O_2	O_3	Model
-	-	-	O_1	1	0	0	
-	-	-	O_3	0	0	1	$M_1 \leftarrow I/p(x_1, x_2, x_3) \rightarrow O/P \{O_1\}$
-	-	-	O_2	0	1	0	
-	-	-	O_1	1	0	0	
-	-	-	O_2	0	1	0	$M_2 \leftarrow I/p(x_1, x_2, x_3) \rightarrow O/P \{O_2\}$
-	-	-	O_3	0	0	1	

$M_3 \leftarrow I/p$

O/P $\{O_3\}$

$\odot \rightarrow M_1 \rightarrow 0.55$
 $\rightarrow M_2 \rightarrow 0.90$
 $\rightarrow M_3 \rightarrow 0.75$

$M_1 \rightarrow \text{train } O_1$

Consider O_1